6	**(a)**	power = VI	C1	
		current = $10.5 \times 103 / 230$	M1	
		= 45.7 A	A0	[2]
	(b) (i)	p.d. across cable = 5.0 V	C1	
		R = 5.0 / 46	C1	
		= 0.11 Ω	A1	[3]
	(ii)	$R = \rho L / A$	C1	
		$0.11 = (1.8 \times 10^{-8} \times 16 \times 2) / A$	C1	
		$A = 5.3 \times 10^{-6}$ m^2	A1	[3]
		(*wires in parallel, not series, allow max 1/3 marks*)		

(c)	**(i)**	*either* power = V^2 / R *or* power $\propto V^2$	C1	
		ratio = $(210 / 230)^2 = 0.83$	A1	[2]
	(ii)	resistance of cable is greater	M1	
		greater power loss/fire hazard/insulation may melt wire may melt/cable gets hot	A1	[2]

7 **(a)** most α-particles deviated through small angles B1